

Finite Element Methods – APE1

Solution Key: Ritz Method for a Clamped Kirchhoff Plate

1. Weak form by two integrations by parts – identification of natural terms

For sake of simplicity, we use $\Delta u = \nabla^2 u$ for the Laplacian. Recall the Kirchhoff operator

$$\Delta(D\Delta u) - = f \quad \text{in } \Omega, \quad (1)$$

where $D(x, y)$ is stiffness coefficient that may that may vary in space.

Step 1: multiply by a virtual variation and integrate. Let δu be an admissible variation. Multiply (1) by δu and integrate over Ω :

$$\int_{\Omega} \Delta(D\Delta u) \delta u \, d\Omega = \int_{\Omega} f \delta u \, d\Omega.$$

Step 2: first integration by parts. Using Green's identity with $w = D\Delta u$ and $v = \delta u$,

$$\int_{\Omega} \Delta w v \, d\Omega = - \int_{\Omega} \nabla w \cdot \nabla v \, d\Omega + \int_{\partial\Omega} \partial_n w v \, ds,$$

we obtain

$$- \int_{\Omega} \nabla(D\Delta u) \cdot \nabla(\delta u) \, d\Omega + \int_{\partial\Omega} \partial_n(D\Delta u) \delta u \, ds = \int_{\Omega} f \delta u \, d\Omega. \quad (2)$$

Step 3: second integration by parts. Apply Green's identity again to the volume term of (2):

$$- \int_{\Omega} \nabla w \cdot \nabla v \, d\Omega = \int_{\Omega} w \Delta v \, d\Omega - \int_{\partial\Omega} w \partial_n v \, ds,$$

with $w = D\Delta u$ and $v = \delta u$. This yields

$$\int_{\Omega} (D\Delta u) \Delta(\delta u) \, d\Omega + \int_{\partial\Omega} \left(\underbrace{\partial_n(D\Delta u)}_{\text{natural term \#1}} \delta u - \underbrace{(D\Delta u)}_{\text{natural term \#2}} \partial_n(\delta u) \right) ds = \int_{\Omega} f \delta u \, d\Omega. \quad (3)$$

Identification with shear force and bending moment. The boundary integral in (3) contains two *natural* quantities:

$$Q_n := - \partial_n(D\Delta u) \quad (\text{Kirchhoff shear force resultant}),$$

$$M_n := - D\Delta u \quad (\text{bending moment resultant}).$$

At this stage, the important point is to understand the principle of the method. The boundary terms obtained after the integrations by parts correspond to the natural boundary quantities (bending moments and Kirchhoff shear forces).

We therefore replace these two terms by their prescribed values on Γ_N . Doing so is equivalent to imposing the natural boundary conditions *weakly* (in the variational sense). In other words, they are not enforced a priori on the admissible space, but are satisfied *a posteriori* through the variational formulation. With these notations, the boundary term becomes

$$\int_{\Gamma_N} \left(-Q_n \delta u + M_n \partial_n(\delta u) \right) ds.$$

Natural boundary conditions. If the boundary is *free*, the virtual fields δu and $\partial_n(\delta u)$ are arbitrary on $\partial\Omega$, hence the coefficients must vanish:

$$M_n = 0 \quad \text{and} \quad Q_n = 0 \quad \text{on } \Gamma_N,$$

i.e.

$$D \Delta u = 0, \quad \partial_n(D \Delta u) = 0 \quad \text{on } \Gamma_N.$$

Clamped boundary conditions. If the boundary is *clamped*, the essential conditions are

$$u = 0, \quad \partial_n u = 0 \quad \text{on } \Gamma_D,$$

so admissible variations satisfy

$$\delta u = 0, \quad \partial_n(\delta u) = 0 \quad \text{on } \Gamma_D,$$

and the entire boundary integral in (3) vanishes automatically.

Weak form (clamped case). For a clamped plate, (3) reduces to: find u such that

$$\int_{\Omega} (D \Delta u) \Delta \delta u \, d\Omega = \int_{\Omega} f \delta u \, d\Omega \quad \forall \delta u \text{ such that } \delta u = \partial_n \delta u = 0 \text{ on } \partial\Omega.$$

Remark. This derivation is intentionally based on the scalar operator $\Delta(D\Delta u)$. The full Kirchhoff–Love model leads to additional (directional) moment components and corner terms; however, the same variational philosophy holds: the boundary terms produced by integration by parts identify the natural data (moment and effective shear) conjugate to $(\partial_n v, v)$.

Setting. Throughout, we consider the square plate

$$\Omega = (0, L) \times (0, L),$$

clamped on its whole boundary:

$$u = 0, \quad \partial_n u = 0 \quad \text{on } \partial\Omega,$$

and subjected to a transverse load $f(x, y)$. For explicit computations in the Ritz part, we will also give the closed-form results for the common case

$$f(x, y) = q \quad (\text{uniform load}), \quad D = \text{constant}.$$

2. Potential Energy.

Assume the (scalar) 4th-order plate model

$$\Delta(D \Delta u) = f \quad \text{in } \Omega,$$

with a boundary split

$$\partial\Omega = \Gamma_D \cup \Gamma_N, \quad \Gamma_D \cap \Gamma_N = \emptyset.$$

On Γ_D (essential boundary conditions) we prescribe

$$u = \bar{u} \quad \text{on } \Gamma_D, \quad \partial_n u = \bar{\theta} \quad \text{on } \Gamma_D.$$

On Γ_N (natural boundary conditions) we prescribe the two natural quantities

$$m := D \Delta u = \bar{m} \quad \text{on } \Gamma_N, \quad q := \partial_n(D \Delta u) = \bar{q} \quad \text{on } \Gamma_N.$$

Admissible set. The minimization is performed over

$$\mathcal{U} = \left\{ u \in H^2(\Omega) : u = \bar{u} \text{ on } \Gamma_D, \partial_n u = \bar{\theta} \text{ on } \Gamma_D \right\}.$$

Potential functional. The functional whose minimizer solves the problem is

$$\Pi(u) = \frac{1}{2} \int_{\Omega} D (\Delta u)^2 d\Omega - \int_{\Omega} f u d\Omega + \int_{\Gamma_N} (\bar{q} u - \bar{m} \partial_n u) ds. \quad (4)$$

Check (first variation). Take variations δu such that $\delta u = \partial_n \delta u = 0$ on Γ_D . Then

$$\delta \Pi(u; \delta u) = \int_{\Omega} D \Delta u \Delta(\delta u) d\Omega - \int_{\Omega} f \delta u d\Omega + \int_{\Gamma_N} (\bar{q} \delta u - \bar{m} \partial_n(\delta u)) ds.$$

Stationarity $\delta \Pi(u; \delta u) = 0$ for all such δu yields the weak form, and after two integrations by parts one recovers

$$\Delta(D \Delta u) = f \text{ in } \Omega, \quad D \Delta u = \bar{m}, \quad \partial_n(D \Delta u) = \bar{q} \text{ on } \Gamma_N,$$

with the essential conditions imposed strongly on Γ_D .

3. What is the “natural” functional space for Kirchhoff plates?

Even without functional analysis, the weak form suggests the key requirement:

$$u_{xx}, u_{yy}, u_{xy} \text{ must be square-integrable over } \Omega.$$

In other words, the admissible functions must have *second (weak) derivatives* in $L^2(\Omega)$. This is precisely the Sobolev space $H^2(\Omega)$.

For a clamped plate, the essential boundary conditions are

$$u = 0, \quad \partial_n u = 0 \quad \text{on } \partial\Omega,$$

so the natural energy space is

$$H_0^2(\Omega) = \left\{ u \in H^2(\Omega) : u = 0 \text{ and } \partial_n u = 0 \text{ on } \partial\Omega \right\},$$

understood in the trace sense.

4. A 1D polynomial $w(x)$ with $w(0) = w(L) = w'(0) = w'(L) = 0$

A simple choice is

$$w(x) = x^2(L - x)^2. \quad (5)$$

Indeed, $w(0) = w(L) = 0$ is obvious, and

$$w'(x) = 2x(L - x)^2 - 2x^2(L - x) = 2x(L - x)((L - x) - x) = 2x(L - x)(L - 2x),$$

so $w'(0) = w'(L) = 0$ as well.

5. Construct an admissible $\phi_1(x, y)$

Define

$$\phi_1(x, y) = w(x)w(y) = x^2(L - x)^2y^2(L - y)^2. \quad (6)$$

Then:

- On $x = 0$ or $x = L$, we have $w(x) = 0$, hence $\phi_1 = 0$. Similarly on $y = 0$ or $y = L$. So $\phi_1 = 0$ on $\partial\Omega$.
- For the normal derivative: on $x = 0$ or $x = L$, the outward normal is $\pm \mathbf{e}_x$, so $\partial_n \phi_1 = \pm \partial_x \phi_1$. But $\partial_x \phi_1 = w'(x)w(y)$ and $w'(0) = w'(L) = 0$, hence $\partial_n \phi_1 = 0$ on those edges. Similarly, on $y = 0$ or $y = L$, $\partial_n \phi_1 = \pm \partial_y \phi_1 = w(x)w'(y) = 0$.

Therefore ϕ_1 satisfies the clamped boundary conditions and is admissible.

6. Ritz with $U_h = \text{span}\{\phi_1\}$ for $\Delta(D\Delta u) = f$

We consider the simplified fourth-order model

$$\Delta(D\Delta u) = f \quad \text{in } \Omega = (0, L)^2,$$

with clamped boundary conditions

$$u = 0, \quad \partial_n u = 0 \quad \text{on } \partial\Omega.$$

Let

$$\phi_1(x, y) = x^2(L - x)^2y^2(L - y)^2$$

as constructed above, and define the one-dimensional trial space

$$U_h = \text{span}\{\phi_1\}.$$

We approximate

$$u_h(x, y) = a \phi_1(x, y),$$

where the scalar coefficient a is found by minimizing the functional

$$\Pi(u) = \frac{1}{2} \int_{\Omega} D(\Delta u)^2 d\Omega - \int_{\Omega} f u d\Omega \quad (\text{clamped: no boundary term}).$$

6.1 Stationarity condition (Ritz equation)

Take a variation $\delta u_h = \delta a \phi_1$. Stationarity of $\Pi(u_h)$ gives

$$\frac{d}{da} \Pi(a\phi_1) = 0 \quad \Longleftrightarrow \quad a \int_{\Omega} D(\Delta \phi_1)^2 d\Omega = \int_{\Omega} f \phi_1 d\Omega.$$

Thus the 1×1 “stiffness” and “load” are

$$K_{11} = \int_{\Omega} D(\Delta \phi_1)^2 d\Omega, \quad F_1 = \int_{\Omega} f \phi_1 d\Omega, \quad a = \frac{F_1}{K_{11}}.$$

6.2 Explicit expression of $\Delta\phi_1$

Write $\phi_1(x, y) = w(x)w(y)$ with $w(x) = x^2(L - x)^2$. Then

$$\Delta\phi_1 = \partial_{xx}\phi_1 + \partial_{yy}\phi_1 = w''(x)w(y) + w(x)w''(y).$$

We compute

$$w(x) = L^2x^2 - 2Lx^3 + x^4, \quad w''(x) = 2L^2 - 12Lx + 12x^2.$$

Hence

$$\Delta\phi_1(x, y) = (2L^2 - 12Lx + 12x^2)w(y) + w(x)(2L^2 - 12Ly + 12y^2). \quad (7)$$

6.3 Definite-integral form of K_{11} and F_1

In full generality (variable $D(x, y)$, generic $f(x, y)$), the Ritz coefficient is

$$a = \frac{\int_0^L \int_0^L f(x, y) w(x)w(y) dx dy}{\int_0^L \int_0^L D(x, y) \left(w''(x)w(y) + w(x)w''(y) \right)^2 dx dy},$$

where w and w'' are given above. This is exactly what one would evaluate symbolically (or numerically) in a CAS.

6.4 Closed form for constant D and uniform load $f = q$ (optional)

Assume

$$D = \text{constant}, \quad f(x, y) = q.$$

Then, using separability of $w(x)w(y)$ and symmetry, one obtains

$$F_1 = \int_{\Omega} q \phi_1 d\Omega = q \left(\int_0^L w(x) dx \right)^2, \quad K_{11} = D \int_{\Omega} (\Delta\phi_1)^2 d\Omega.$$

With

$$\int_0^L w(x) dx = \int_0^L x^2(L - x)^2 dx = \frac{L^5}{30},$$

we get

$$F_1 = q \frac{L^{10}}{900}.$$

Moreover, letting

$$I_0 = \int_0^L w(x)^2 dx = \frac{L^9}{630}, \quad I_2 = \int_0^L (w''(x))^2 dx = \frac{4}{5}L^5, \quad I_1 = \int_0^L w(x)w''(x) dx = -\frac{2}{15}L^7,$$

and using

$$\int_{\Omega} (\Delta\phi_1)^2 = 2 I_2 I_0 + 2 I_1^2,$$

we obtain

$$\int_{\Omega} (\Delta\phi_1)^2 d\Omega = \frac{4}{75}L^{14}, \quad K_{11} = D \frac{4}{75}L^{14}.$$

Therefore

$$a = \frac{F_1}{K_{11}} = \frac{qL^{10}/900}{D(4L^{14}/75)} = \frac{1}{48} \frac{q}{DL^4}.$$

The one-term Ritz approximation is thus

$$u_h(x, y) = \frac{1}{48} \frac{q}{D L^4} x^2 (L - x)^2 y^2 (L - y)^2.$$

Remark. Since U_h is one-dimensional, this approximation is the best (in the energy sense induced by $\int_{\Omega} D(\Delta \cdot)^2$) among all multiples of ϕ_1 . Adding more admissible basis functions can only decrease the energy and the error in that norm.

7. Admissibility of $\phi_2 = \phi_1(x(L - x) + y(L - y))$

Let

$$s(x, y) = x(L - x) + y(L - y), \quad \phi_2 = \phi_1 s.$$

- On $\partial\Omega$, $\phi_1 = 0$, hence $\phi_2 = 0$.
- On $x = 0$ or $x = L$,

$$\partial_x \phi_2 = (\partial_x \phi_1) s + \phi_1 (\partial_x s).$$

But on $x = 0$ or $x = L$, we have $\phi_1 = 0$ and also $\partial_x \phi_1 = w'(x)w(y) = 0$, so $\partial_x \phi_2 = 0$ there, hence $\partial_n \phi_2 = 0$ on these edges. Similarly on $y = 0$ or $y = L$.

Therefore ϕ_2 is admissible (clamped on all edges).

8. Why ϕ_1 and ϕ_2 are linearly independent

Assume $\alpha\phi_1 + \beta\phi_2 = 0$ in Ω . Factor ϕ_1 :

$$\phi_1(\alpha + \beta s(x, y)) = 0 \quad \text{in } \Omega.$$

Inside the open domain $(0, L)^2$, $\phi_1 > 0$ (it is a product of strictly positive terms), so we must have

$$\alpha + \beta s(x, y) = 0 \quad \forall (x, y) \in (0, L)^2.$$

But $s(x, y) = x(L - x) + y(L - y)$ is *not constant* in the domain (it varies with x and y), hence the only possibility is $\beta = 0$ and then $\alpha = 0$. Thus $\{\phi_1, \phi_2\}$ is linearly independent.

8. Ritz with $U_h = \text{span}\{\phi_1, \phi_2\}$ for $\Delta(D\Delta u) = f$

Here every computation was done symbolically, not by hand!

We keep the simplified fourth-order model

$$\Delta(D\Delta u) = f \quad \text{in } \Omega = (0, L)^2, \quad u = 0, \quad \partial_n u = 0 \quad \text{on } \partial\Omega,$$

and we use the two admissible trial functions

$$\phi_1(x, y) = w(x)w(y), \quad \phi_2(x, y) = \phi_1(x, y) s(x, y),$$

with

$$w(x) = x^2(L - x)^2, \quad s(x, y) = x(L - x) + y(L - y).$$

We consider the two-dimensional space

$$U_h = \text{span}\{\phi_1, \phi_2\}$$

and the Ritz ansatz

$$u_h(x, y) = a_1\phi_1(x, y) + a_2\phi_2(x, y).$$

8.1 Functional and Ritz system

For clamped boundary conditions, the functional is

$$\Pi(u) = \frac{1}{2} \int_{\Omega} D (\Delta u)^2 d\Omega - \int_{\Omega} f u d\Omega.$$

Stationarity of Π over U_h yields the 2×2 linear system

$$\sum_{j=1}^2 K_{ij} a_j = F_i, \quad i = 1, 2,$$

with

$$\boxed{K_{ij} = \int_{\Omega} D (\Delta \phi_i) (\Delta \phi_j) d\Omega, \quad F_i = \int_{\Omega} f \phi_i d\Omega.} \quad (8)$$

Equivalently, the weak form reads: find $u_h \in U_h$ such that

$$\int_{\Omega} D \Delta u_h \Delta v_h d\Omega = \int_{\Omega} f v_h d\Omega \quad \forall v_h \in U_h,$$

which leads to the same matrix entries (8).

8.2 Laplacians of ϕ_1 and ϕ_2

Since $\phi_1 = w(x)w(y)$,

$$\Delta \phi_1 = w''(x)w(y) + w(x)w''(y), \quad w''(x) = 2L^2 - 12Lx + 12x^2.$$

For $\phi_2 = \phi_1 s$ with $s = x(L - x) + y(L - y)$, use the product rule

$$\Delta(\phi_1 s) = s \Delta \phi_1 + \phi_1 \Delta s + 2 \nabla \phi_1 \cdot \nabla s.$$

Now

$$\nabla s = (L - 2x, L - 2y), \quad \Delta s = -4,$$

and

$$\nabla \phi_1 = (w'(x)w(y), w(x)w'(y)), \quad w'(x) = 2x(L - x)(L - 2x).$$

Hence

$$\boxed{\Delta \phi_2 = s \Delta \phi_1 - 4 \phi_1 + 2 w'(x)w(y) (L - 2x) + 2 w(x)w'(y) (L - 2y).} \quad (9)$$

8.3 Definite-integral expressions (what students must be able to write)

Plug (8) together with $\Delta \phi_1$ and $\Delta \phi_2$ from (9) into the integrals:

$$K_{11} = \int_0^L \int_0^L D(x, y) (\Delta \phi_1(x, y))^2 dx dy,$$

$$K_{12} = K_{21} = \int_0^L \int_0^L D(x, y) \Delta \phi_1(x, y) \Delta \phi_2(x, y) dx dy,$$

$$K_{22} = \int_0^L \int_0^L D(x, y) (\Delta \phi_2(x, y))^2 dx dy,$$

and

$$F_1 = \int_0^L \int_0^L f(x, y) \phi_1(x, y) dx dy, \quad F_2 = \int_0^L \int_0^L f(x, y) \phi_2(x, y) dx dy.$$

Solve the 2×2 system to obtain (a_1, a_2) .

8.4 Closed form for constant D and uniform load $f = q$ (optional)

Assume D is constant and $f = q$. Then

$$F_1 = q \left(\int_0^L w(x) dx \right)^2 = \frac{qL^{10}}{900}.$$

Moreover,

$$F_2 = q \int_{\Omega} \phi_1 s d\Omega = q \left[\left(\int_0^L w(x) x(L-x) dx \right) \left(\int_0^L w(y) dy \right) + \left(\int_0^L w(x) dx \right) \left(\int_0^L w(y) y(L-y) dy \right) \right] = \frac{q}{2}$$

since $\int_0^L w(x)x(L-x) dx = L^7/140$ and $\int_0^L w(x) dx = L^5/30$.

The stiffness entries are still given by (8) with D factored out. A symbolic computation (CAS) gives

$$K_{11} = D \frac{4}{75} L^{14}, \quad K_{12} = D \frac{52}{1575} L^{16}, \quad K_{22} = D \frac{2368}{363825} L^{18}.$$

Therefore

$$\begin{bmatrix} \frac{4}{75} L^{14} & \frac{52}{1575} L^{16} \\ \frac{52}{1575} L^{16} & \frac{2368}{363825} L^{18} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{q}{D} \begin{bmatrix} \frac{L^{10}}{900} \\ \frac{L^{12}}{2100} \end{bmatrix}.$$

Solving yields

$$a_1 = \frac{1183}{3624} \frac{q}{D L^4}, \quad a_2 = -\frac{1925}{9664} \frac{q}{D L^6}.$$

Hence the two-term Ritz approximation is

$$u_h(x, y) = \frac{1183}{3624} \frac{q}{D L^4} \phi_1(x, y) - \frac{1925}{9664} \frac{q}{D L^6} \phi_2(x, y).$$

Remark (energy improvement). Since $\text{span}\{\phi_1\} \subset \text{span}\{\phi_1, \phi_2\}$, the Ritz solution with two basis functions minimizes Π over a larger space, hence

$$\Pi(u_h^{(2)}) \leq \Pi(u_h^{(1)}),$$

and the error decreases in the energy norm $\|v\|_E^2 = \int_{\Omega} D(\Delta v)^2 d\Omega$.

10. Analytical solution (for $D\Delta^2 u = q$) and comparison at the center

In this section we consider the constant-coefficient case

$$D\Delta^2 u = q \quad \text{in } \Omega = (0, L)^2, \quad u = 0, \quad \partial_n u = 0 \quad \text{on } \partial\Omega,$$

(i.e. $\Delta(D\Delta u) = q$ with constant D), with a *uniform load* q .

Analytical solution. The exact solution can be written as a (rapidly convergent) series obtained by separation of variables. In particular, the center deflection has the universal form

$$u_{\text{exact}}\left(\frac{L}{2}, \frac{L}{2}\right) = -k_{\text{clamped}} \frac{qL^4}{D}, \quad k_{\text{clamped}} \approx 1.26 \times 10^{-3}. \quad (10)$$

Timoshenko's book on plates [1] is by far the most important reference on that topic.

Center value of the 1-term Ritz approximation

With $u_h^{(1)} = a \phi_1$ and

$$a = \frac{1}{48} \frac{q}{DL^4}, \quad \phi_1 = w(x)w(y), \quad w(x) = x^2(L-x)^2,$$

we have

$$\phi_1\left(\frac{L}{2}, \frac{L}{2}\right) = \left(\frac{L^4}{16}\right)^2 = \frac{L^8}{256},$$

hence

$$u_h^{(1)}\left(\frac{L}{2}, \frac{L}{2}\right) = \frac{1}{48} \frac{q}{DL^4} \cdot \frac{L^8}{256} = \frac{1}{12288} \frac{qL^4}{D} \approx 8.14 \times 10^{-5} \frac{qL^4}{D}. \quad (11)$$

Center value of the 2-term Ritz approximation

With $u_h^{(2)} = a_1 \phi_1 + a_2 \phi_2$ and

$$\phi_2 = \phi_1 s, \quad s = x(L-x) + y(L-y),$$

(using the coefficients found previously for the uniform-load, constant- D case)

$$a_1 = \frac{1183}{3624} \frac{q}{DL^4}, \quad a_2 = -\frac{1925}{9664} \frac{q}{DL^6},$$

we use

$$s\left(\frac{L}{2}, \frac{L}{2}\right) = \frac{L^2}{2}, \quad \phi_2\left(\frac{L}{2}, \frac{L}{2}\right) = \frac{L^{10}}{512}.$$

Therefore

$$u_h^{(2)}\left(\frac{L}{2}, \frac{L}{2}\right) = \left(\frac{13153}{14843904}\right) \frac{qL^4}{D} \approx 8.86 \times 10^{-4} \frac{qL^4}{D}. \quad (12)$$

Comparison with the analytical center value

From (10)–(12), comparing magnitudes,

$$\frac{|u_h^{(1)}(L/2, L/2)|}{|u_{\text{exact}}(L/2, L/2)|} \approx \frac{8.14 \times 10^{-5}}{1.26 \times 10^{-3}} \approx 6.5\%, \quad \frac{|u_h^{(2)}(L/2, L/2)|}{|u_{\text{exact}}(L/2, L/2)|} \approx \frac{8.86 \times 10^{-4}}{1.26 \times 10^{-3}} \approx 70\%.$$

So adding the second admissible basis function improves the center deflection by more than one order of magnitude in relative accuracy.

11. In what mathematical sense is the two-function Ritz approximation better?

Let $U_1 = \text{span}\{\phi_1\}$ and $U_2 = \text{span}\{\phi_1, \phi_2\}$. Then $U_1 \subset U_2 \subset H_0^2(\Omega)$.

Energy monotonicity. Because $u_h^{(k)}$ is the minimizer of Π over U_k ,

$$\Pi(u_h^{(2)}) \leq \Pi(u_h^{(1)}) \leq \Pi(v) \quad \forall v \in U_1.$$

So the two-dimensional approximation *cannot be worse* in terms of the minimized potential energy.

Best approximation in the energy norm. Define the *energy norm* induced by $a(\cdot, \cdot)$:

$$\|v\|_E^2 := a(v, v).$$

For the exact solution u and the Ritz solution $u_h^{(k)} \in U_k$, one has the *Galerkin orthogonality*

$$a(u - u_h^{(k)}, v_h) = 0 \quad \forall v_h \in U_k,$$

and the *best-approximation property*

$$\|u - u_h^{(k)}\|_E = \min_{v_h \in U_k} \|u - v_h\|_E.$$

Since $U_1 \subset U_2$, it follows immediately that

$$\|u - u_h^{(2)}\|_E \leq \|u - u_h^{(1)}\|_E.$$

This is the precise mathematical meaning in which the two-function Ritz approximation is better: it yields a smaller (or equal) error in the energy norm, and a lower (or equal) total potential energy.

12: The full analytical solution

Me: Develop the full analytical solution for the clamped plate problem $(0, L) \times (0, L)$ with a uniform unit load.

AI: We consider the classical constant-coefficient problem

$$D \Delta^2 u = q \quad \text{in } \Omega = (0, L) \times (0, L), \quad u = 0, \quad \partial_n u = 0 \quad \text{on } \partial\Omega, \quad (13)$$

with a uniform transverse load q and constant rigidity D .

Important remark (why the analytical solution is not “simple”). For simply supported plates, a pure double-sine (Navier) series diagonalizes Δ^2 . For a *fully clamped* rectangle, enforcing both $u = 0$ and $\partial_n u = 0$ on all four edges prevents such a straightforward diagonalization: one still obtains an exact series solution, but its coefficients are determined by solving an infinite (rapidly convergent) linear system. This is why tables and reference values are commonly used in engineering practice.

12.1 Symmetry reduction

Because the load is uniform and the domain is a square, the solution is symmetric with respect to the midlines. In particular,

$$u(x, y) = u(L - x, y) = u(x, L - y) = u(L - x, L - y),$$

and the maximum deflection occurs at the center $(L/2, L/2)$.

12.2 Lévy-type series representation (exact)

A standard exact approach is to expand in a trigonometric series in one direction and solve the resulting fourth-order ODEs in the other direction (Lévy method). One convenient (and widely used) representation for the clamped square under uniform load writes the solution as

$$u(x, y) = \sum_{m=1,3,5,\dots}^{\infty} U_m(y) \sin\left(\frac{m\pi x}{L}\right), \quad (14)$$

where only odd indices appear because the right-hand side is constant. For each m , inserting (14) into (13) gives the ODE

$$U_m''''(y) - 2\alpha_m^2 U_m''(y) + \alpha_m^4 U_m(y) = \frac{4q}{D} \frac{1}{m\pi} \frac{1}{\alpha_m^0} \frac{1}{\pi} \frac{1}{m} \quad \text{with} \quad \alpha_m = \frac{m\pi}{L}. \quad (15)$$

More precisely, since

$$q = \sum_{m=1,3,5,\dots}^{\infty} \frac{4q}{m\pi} \sin\left(\frac{m\pi x}{L}\right),$$

the forcing term for mode m is $(4q)/(m\pi)$. Equation (15) can thus be written as

$$U_m'''' - 2\alpha_m^2 U_m'' + \alpha_m^4 U_m = \frac{4q}{D} \frac{1}{m\pi}. \quad (16)$$

General solution for each mode. A particular solution of (16) is the constant

$$U_m^{(p)} = \frac{4q}{D} \frac{1}{m\pi} \frac{1}{\alpha_m^4}.$$

The homogeneous solution solves $(\partial_y^2 - \alpha_m^2)^2 U = 0$, hence

$$U_m^{(h)}(y) = (A_m + B_m y) \cosh(\alpha_m y) + (C_m + D_m y) \sinh(\alpha_m y).$$

Therefore

$$U_m(y) = \frac{4q}{D} \frac{1}{m\pi} \frac{1}{\alpha_m^4} + (A_m + B_m y) \cosh(\alpha_m y) + (C_m + D_m y) \sinh(\alpha_m y). \quad (17)$$

12.3 Enforcing the clamped boundary conditions

The representation (14) enforces $u(0, y) = u(L, y) = 0$ automatically because of the sine factor. The remaining clamped conditions are:

$$\partial_x u(0, y) = 0, \quad \partial_x u(L, y) = 0 \quad \text{and} \quad u(x, 0) = u(x, L) = 0, \quad \partial_y u(x, 0) = \partial_y u(x, L) = 0.$$

Key point: for a clamped boundary, the conditions at $x = 0$ and $x = L$ involve $\partial_x u$, which produces cosines rather than sines. As a result, the coefficients A_m, B_m, C_m, D_m in (17) are *not* obtained independently for each m ; instead, enforcing $\partial_x u = 0$ on the vertical edges couples the modes and leads to an infinite linear system.

Concretely, one proceeds as follows:

- Impose $u(x, 0) = u(x, L) = 0$ and $\partial_y u(x, 0) = \partial_y u(x, L) = 0$ to express (A_m, B_m, C_m, D_m) in terms of *one* unknown amplitude per mode. This step is purely algebraic, using (17).
- Then enforce $\partial_x u(0, y) = \partial_x u(L, y) = 0$ in the sense of Fourier coefficients: multiply by $\sin(n\pi y/L)$ and integrate in y to obtain a linear system for the remaining unknown modal amplitudes.
- Truncating to $m = 1, 3, \dots, 2N-1$ yields an $N \times N$ system whose solution converges rapidly as $N \rightarrow \infty$.

This procedure yields an *exact* analytical series solution (the only numerical step is the solution of a well-conditioned linear system for the series coefficients).

12.4 Center deflection and the constant k_{clamped}

The exact center deflection can be written in the dimensionless form

$$u\left(\frac{L}{2}, \frac{L}{2}\right) = -k_{\text{clamped}} \frac{qL^4}{D}, \quad (18)$$

where the constant k_{clamped} is obtained by evaluating the converged series solution at $(L/2, L/2)$. For a fully clamped square plate under uniform load, the classical tabulated value is

$$k_{\text{clamped}} \approx 1.26 \times 10^{-3}. \quad (19)$$

References

- [1] S. Timoshenko and S. Woinowsky–Krieger, *Theory of Plates and Shells*, 2nd ed., McGraw–Hill, 1959. (See the tables of deflections for a clamped square plate under uniform load.)
- [2] W. C. Young and R. G. Budynas, *Roark’s Formulas for Stress and Strain*, 7th ed., McGraw–Hill, 2002. (Table: rectangular plates, all edges fixed.)